Inorganic and Materials Cumulative Exam
2/28/2025

Answer any 5 of the following 10 questions (20 points each). Any answers provided beyond the requested five will not be graded. Remember: Your answers to these questions are meant to provide evidence of broad, graduate-school level chemistry knowledge. Consider this in your choice of terminology and examples as answers to these questions.

1. Define hydrate isomerism and provide an example of a set of compounds to explain this isomerization. Draw diagrams to support your answer.

2. (a) (12 pts) Arrange the following complex octahedral ions (in aqueous solution) in order of increasing absorption wavelength and explain your rationale: $[Ti(H_2O)_6]^{3+}$,$[TiF_6]^{3-}$, $[Ti(CO)_6]^{3+}$,$[Ti(NH_3)_6]^{3+}$

(b) (8 pts) Which has a higher characteristic absorption wavelength: $[V(H_2O)_6]^{2+}$ or $[V(H_2O)_6]^{3+}$? Explain your rationale.

3. (a) (4 pts) What is the Lewis structure for the square planar (cis)-$PtBr_2Cl_2$?

(b) (4 pts) What is the point group of the compound?

(c) (6 pts) Using the included character table, how many IR-active modes does (cis)-$PtBr_2Cl_2$ have?

(d) (6 pts) Using the included character table, how many Raman-active modes does (cis)-$PtBr_2Cl_2$ have?

4. Consider LiF as a molecule to predict some of its electronic structure.

(a) (12 pts) Using a molecular orbital diagram, describe the electronic structure of LiF (Li 1s orbital potential = -56eV, Li 2s orbital potential = -5.4eV, F 2s orbital potential = -40.2eV, F 2p orbital potential = -18.7eV).

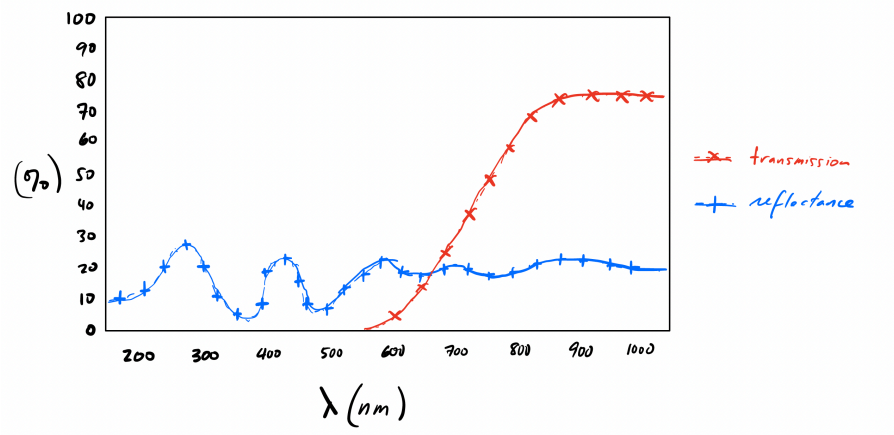
(b) (8 pts) In nature, LiF is a solid rather than a molecule. Using the MO diagram from (a) why is LiF not a molecule?

5. In XPS, the Ti ($2p_{3/2}$) binding energy exhibits the following trend:

Solid/compound	Ti(s)	TiO(s)	TiO_2(s)
Binding Energy $2p_{3/2}$ (eV)	453.9	455.2	458.7

(a) (12 pts) Why is this the case? Please explain this chemical shift in qualitative terms with respect to the local electronic environment around the Ti atoms and their oxidation state in each solid.

(b) (8 pts) Predict the Ti ($2p_{3/2}$) binding energy for TiS_2. Provide an explanation for your prediction.



6. Pretend you performed UV-Vis spectroscopy on a semiconductor thin film sample deposited on a transparent substrate (e.g., quartz or sapphire). Use the simulated transmission and reflectance data (above) to answer the following questions:

(a) (5 pts) How would you calculate the absorption spectra from these measurements?

(b) (7 pts) What is the band gap of the semiconductor? Explain your answer.

(c) (8 pts) What color is the film? Explain your answer.

7. (4 pts each) Assume you are given a solid sample of unknown composition. Describe one technique for each of (a-e) that you would use to unambiguously determine the following properties of the material experimentally. Please provide a brief description of how the measurement would determine the properties.

(a) Atomic structure and symmetry

(b) Composition

(c) Electrical properties (insulator vs. conductor vs. semiconductor)

(d) Band gap

(e) Band structure

8. Describe the operation of a water electrolysis reactor. Sketch a diagram of the reactor and label important components, including one example of specific materials used in each.

9. (a) (10 pts) How is bonding in an ionic solid different than bonding in a network covalent solid? Include at least 2 specific examples of differences.

(b) (10 pts) How is bonding in a metal different than bonding in a network covalent solid? Include at least 2 specific examples of differences.

10. (a) (7 pts) Describe the basic operating procedure of a Raman spectrometer, including an example of the main features of a Raman spectrum.

(b) (8 pts) An enhancement in Raman absorption signal is observed for molecules on a rough gold surface, a surface-enhanced Raman scattering (SERS) effect. Why is Pt not a good candidate for this SERS in a system where Au IS a good candidate?

(c) (5 pts) Copper can have a similar SERS effect, but is not used for practical reasons, particularly in aqueous environments. Why is this?

Some useful character tables

C_2	E	C_2		
A	1	1	z, R_z	x^2, y^2, z^2, xy
B	1	-1	x, y, R_x, R_y	yz, xz

C_{2v}	E	C_2	$\sigma_v(xz)$	$\sigma_v'(yz)$		
A_1	1	1	1	1	z	x^2, y^2, z^2
A_2	1	1	-1	-1	R_z	xy
B_1	1	-1	1	-1	x, R_y	xz
B_2	1	-1	-1	1	y, R_x	yz

C_{3v}	E	$2C_3$	$3\sigma_v$		
A_1	1	1	1	z	$x^2 + y^2, z^2$
A_2	1	1	-1	R_z	
E	2	-1	0	$(x, y), (R_x, R_y)$	$(x^2 - y^2, xy), (xz, yz)$

C_{4v}	E	$2C_4$	C_2	$2\sigma_v$	$2\sigma_d$		
A_1	1	1	1	1	1	z	$x^2 + y^2, z^2$
A_2	1	1	1	-1	-1	R_z	
B_1	1	-1	1	1	-1		$x^2 - y^2$
B_2	1	-1	1	-1	1		xy
E	2	0	-2	0	0	$(x, y), (R_x, R_y)$	(xz, yz)

D_2	E	$C_2(z)$	$C_2(y)$	$C_2(x)$		
A	1	1	1	1		x^2, y^2, z^2
B_1	1	1	-1	-1	z, R_z	xy
B_2	1	-1	1	-1	y, R_y	xz
B_3	1	-1	-1	1	x, R_x	yz

D_3	E	$2C_3$	$3C_2$		
A_1	1	1	1		$x^2 + y^2, z^2$
A_2	1	1	-1	z, R_z	
E	2	-1	0	$(x, y), (R_x, R_y)$	$(x^2 - y^2, xy), (xz, yz)$

D_{2d}	E	$2S_4$	C_2	$2C_2'$	$2\sigma_d$		
A_1	1	1	1	1	1		$x^2 + y^2, z^2$
A_2	1	1	1	-1	-1	R_z	
B_1	1	-1	1	1	-1		$x^2 - y^2$
B_2	1	-1	1	-1	1	z	xy
E	2	0	-2	0	0	$(x, y), (R_x, R_y)$	(xz, yz)

D_{3d}	E	$2C_3$	$3C_2$	i	$2S_6$	$3\sigma_d$		
A_{1g}	1	1	1	1	1	1		$x^2 + y^2, z^2$
A_{2g}	1	1	-1	1	1	-1	R_z	
E_g	2	-1	0	2	-1	0	(R_x, R_y)	$(x^2 - y^2, xy)$ (xz, yz)
A_{1u}	1	1	1	-1	-1	-1		
A_{2u}	1	1	-1	-1	-1	1	z	
E_u	2	-1	0	-2	1	0	(x, y)	

D_{2h}	E	$C_2(z)$	$C_2(y)$	$C_2(x)$	i	$\sigma(xy)$	$\sigma(xz)$	$\sigma(yz)$		
A_g	1	1	1	1	1	1	1	1		x^2, y^2, z^2
B_{1g}	1	1	−1	−1	1	1	−1	−1	R_z	xy
B_{2g}	1	−1	1	−1	1	−1	1	−1	R_y	xz
B_{3g}	1	−1	−1	1	1	−1	−1	1	R_x	yz
A_u	1	1	1	1	−1	−1	−1	−1		
B_{1u}	1	1	−1	−1	−1	−1	1	1	z	
B_{2u}	1	−1	1	−1	−1	1	−1	1	y	
B_{3u}	1	−1	−1	1	−1	1	1	−1	x	

D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$		
A_1'	1	1	1	1	1	1		$x^2 + y^2, z^2$
A_2'	1	1	−1	1	1	−1	R_z	
E'	2	−1	0	2	−1	0	(x, y)	$(x^2 - y^2, xy)$
A_1''	1	1	1	−1	−1	−1		
A_2''	1	1	−1	−1	−1	1	z	
E''	2	−1	0	−2	1	0	(R_x, R_y)	(xz, yz)

D_{4h}	E	$2C_4$	C_2	$2C_2'$	$2C_2''$	i	$2S_4$	σ_h	$2\sigma_v$	$2\sigma_d$		
A_{1g}	1	1	1	1	1	1	1	1	1	1		$x^2 + y^2, z^2$
A_{2g}	1	1	1	−1	−1	1	1	1	−1	−1	R_z	
B_{1g}	1	−1	1	1	−1	1	−1	1	1	−1		$x^2 - y^2$
B_{2g}	1	−1	1	−1	1	1	−1	1	−1	1		xy
E_g	2	0	−2	0	0	2	0	−2	0	0	(R_x, R_y)	(xz, yz)
A_{1u}	1	1	1	1	1	−1	−1	−1	−1	−1		
A_{2u}	1	1	1	−1	−1	−1	−1	−1	1	1	z	
B_{1u}	1	−1	1	1	−1	−1	1	−1	−1	1		
B_{2u}	1	−1	1	−1	1	−1	1	−1	1	−1		
E_u	2	0	−2	0	0	−2	0	2	0	0	(x, y)	